About this course

Welcome, what to expect, evaluation criteria

19 February 2026

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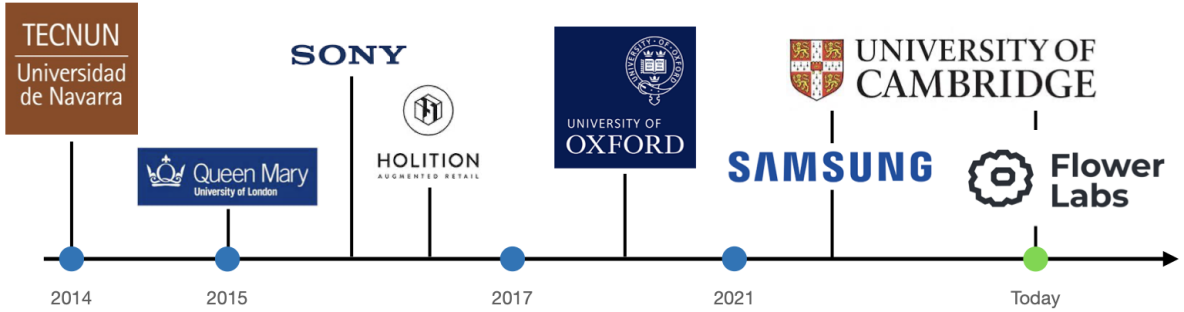
Federated AI, efficient ML, embedded ML



What to expect for this course?

- Gain a strong foundation in Federated AI → Often the only way to create/use AI with private data
- Design and implement you own Federated AI models → *Using Flower, an open source framework*
- Learn to identify real-world challenges of Federated AI Systems → *Comms, privacy, data heterogeneity*
- Measure compute/memory footprint of AI models
- Overview of latest research in Federate AI and AI Systems → 15+ landmark & recent papers
- 3x Practicals in ~~Colab~~ VSCode
- Time allocation: 40% slides 60% coding exercises → Learning by doing !!
- Interactive class (ask questions!) → Ask me anything in English or Spanish :)
- Taught in English

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Evaluation

- **Practical A:** Flower basics: 0%
- **Practical B:** Creating a custom Flower App: 0%
- **Practical C:** Study Data heterogeneity in Federated AI: 50% (code + short report)
- **Final Project:** … more details tomorrow … : 50% (code + short report)

vast.ai + Flower A Friendly Federated Learning Framework + PyTorch

Schedule

Today

- Introduction
- Introduction to Federated Learning
- Introduction to Flower
- Vast.ai Walkthrough
- Federated AI Practical

Tomorrow

- Advanced topics in Federated Learning
- Federated AI Practical
- Project Design
- Continue with Practical or start with Project

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